# Problem D. Directed Acyclic Graph

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|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 512 megabytes |

Given two integers $n$ and $m$, you need to construct a directed acyclic graph (DAG) $G = (V, E)$ with exactly $n$ vertices and $m$ edges.

In the graph $G$, a vertex $v$ is called *reachable* from vertex $u$ if and only if there exists a path in the graph that starts at vertex $u$ and ends at vertex $v$.

For a non-empty set of vertices $A \subseteq V$, a vertex $w$ is defined as *good* for $A$ if and only if it is reachable from **every** vertex in $A$, and we denote $f(A)$ as the set of all good vertices for $A$.

The graph you constructed should satisfy both of the following constraints:

- For every vertex $i$ ($1 \le i \le n$), it is reachable from vertex 1.
- There exist $k$ distinct non-empty sets of vertices $S_1, S_2, \ldots, S_k$, such that $f(S_1), f(S_2), \ldots, f(S_k)$ are pairwise distinct. Note that $f(S_i)$ can be empty.

To prove the graph $G$ you constructed satisfies the second constraint, you also need to provide $k$ sets $S_1, S_2, \ldots, S_k$ that satisfy the second constraint.

## Input

The only line of the input contains three integers $n, m$, and $k$.

There are only 2 tests in this problem:

1. $n = 5$, $m = 6$, $k = 6$;
2. $n = 100$, $m = 128$, $k = 16\,000$.

## Output

The first $m$ lines of the output describe the graph $G$ you construct. Each line contains two integers $u, v$ representing an edge from $u$ to $v$ in the graph.

The next $k$ lines of the output describe the $k$ sets of vertices you provide. The $i$-th line first contains the size of the set $S_i$, followed by the $|S_i|$ numbers representing each vertex in the set.

## Example

| standard input | standard output |
|---|---|
| 5 6 6 | 1 2<br>1 3<br>2 4<br>3 5<br>2 5<br>3 4<br>1 1<br>1 2<br>1 3<br>1 4<br>1 5<br>2 2 3 |

## Note

In the example, the output constructs a graph with $n = 5$ vertices and $m = 6$ edges. The corresponding $k = 6$ sets are $S_1 = \{1\}, S_2 = \{2\}, S_3 = \{3\}, S_4 = \{4\}, S_5 = \{5\}, S_6 = \{2, 3\}$.

Here, vertices 4 and 5 can both be reached from any element in $S_6 = \{2, 3\}$, so $f(\{2, 3\}) = \{4, 5\}$. Together with $f(S_1) = \{1, 2, 3, 4, 5\}, f(S_2) = \{2, 4, 5\}, f(S_3) = \{3, 4, 5\}, f(S_4) = \{4\}, f(S_5) = \{5\}$, these sets are all distinct, satisfying the constraints.

